

SystemOS — Habit Tracker & Digital Wellness Monitor

An On-Device AI Powered Student Performance & Behavioral Analytics System

Domain	Habit Tracking & Digital Health	Architecture	Next.js 16 + Local LLM (Ollama)	Submission	Official Project Report
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1. Problem Statement & Context

Core Challenge: High academic and professional competition demands extreme daily consistency, yet hyper-connected digital environments subject students and workers to severe distraction overload, burnout, and privacy vulnerabilities in cloud habit apps.

In today’s modern world, competition is omnipresent. Every sector—from competitive examinations (JEE, NEET, Civil Services) to corporate technology roles—is saturated with qualified candidates. While everyone aspires to achieve a successful life, getting there requires long, sustained, and hard work paired with unwavering daily consistency.

Unfortunately, in this modern era, the abundance of knowledge at our fingertips is accompanied by an unprecedented volume of distractive sources. Social media, constant notifications, and entertainment algorithms fragment human attention spans, making long-term habit formation extremely difficult.

A. Key Issues Tackled

- Cognitive Friction & Distraction Overload:** Constant digital interruptions break study flow and prevent deep work habits from taking root.
- The Consistency Deficit & Burnout:** Most individuals start with high motivation but drop off within 14 days due to a lack of feedback mechanisms and unmanaged stress.
- Passive Tracking Without Diagnosis:** Conventional habit tools only log missed checkmarks. They fail to find mistakes in oneself, analyze patterns, or provide actionable feedback for self-correction.
- Privacy Compromises in Cloud Applications:** Existing wellness monitors upload sensitive user habits, stress logs, and personal health metrics to third-party cloud servers.

B. Urgency & Societal Relevance

Helping individuals reduce unnecessary friction, identify personal flaws, and build positive behavioral momentum directly improves productivity, mental health, and societal contribution. SystemOS converts habit tracking from a passive checklist into an active, privacy-first diagnostic operating system.

2. Ideation & Approach

Our project began by researching popular habit tracking applications inspired by platforms across Instagram ads, YouTube videos, and productivity blogs. While these apps offered initial inspiration, every existing tool presented significant caveats:

Existing App Pattern	Key Caveat / Drawback	SystemOS Solution & Approach
Cloud API AI Chatbots	Uploads personal health logs; expensive cloud subscriptions	Local Ollama LLM (qwen3:4b) — 100% private on-device execution
Simple Daily Checkboxes	No month-scoped boundaries; allows manipulating future dates	Future-Date Protection Engine & Month-Scoped 35-Day Grid
Isolated Text Lists	Lacks analytical visual correlations between habits & wellness	Analytical Graphs (Radar Chart, Heatmap, Health Score Gauges)
Unregulated Goals	Allows invalid habit goals (e.g. setting 35 days in 30-day month)	Dynamic Goal Clamping (1–31 Days) scoped to active month

A. Our Technical Rationale & Key Requirements

To build a frictionless habit tracking system, we established 5 essential requirements:

- 1. **Analytical Graphs & Radial Metrics:** Recharts SVG Radar charts, 365-day heatmaps, and Area trend graphs to visualize mistakes and growth.
- 2. **Sleek Single-View GUI:** Japandi Soft Stone & Deep Space Navy Dark Mode design system showing metrics in a unified dashboard without tedious scrolling.
- 3. **Social Leaderboard & Peer Motivation:** Healthy competition and status badges to induce accountability and constructive regret over bad habits.
- 4. **Gamified Reward System:** Level progression (10 tiers), 15 unlockable badges, coin economy, and confetti celebrations to make consistency rewarding.
- 5. **Streak & Failure Prediction System:** Multi-month streak tracking paired with a 14-day rolling failure prediction algorithm to detect slumps before habits fail.

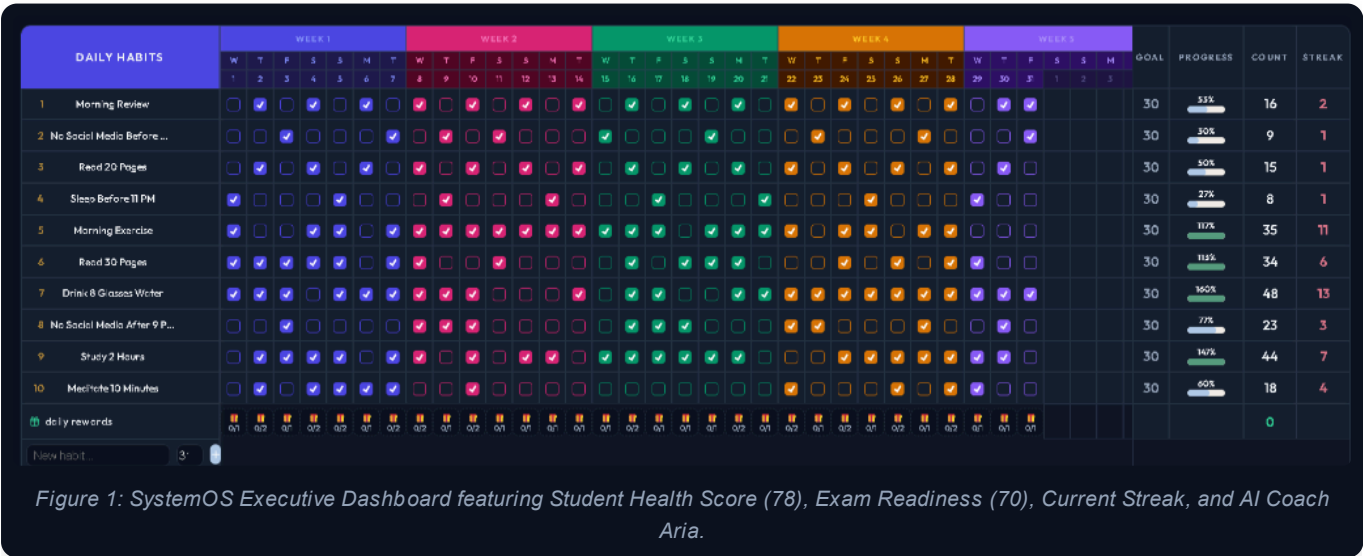


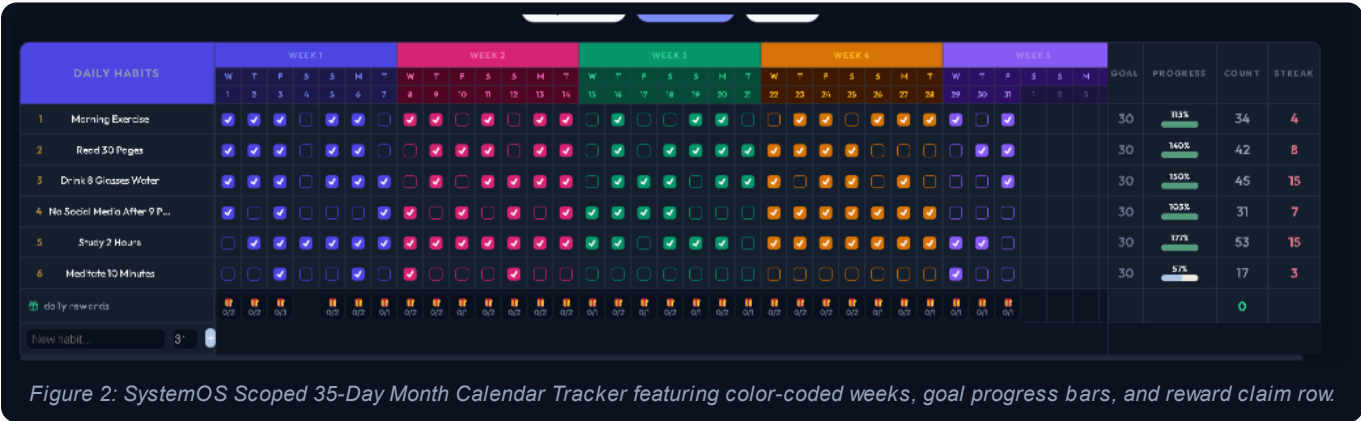
Figure 1: SystemOS Executive Dashboard featuring Student Health Score (78), Exam Readiness (70), Current Streak, and AI Coach Aria.

3. Functionality & Key Features

SystemOS is structured into 7 core functional modules providing a complete walkthrough of student productivity, focus, and digital wellness.

A. System Modules & Functional Walkthrough

Module Name	Functional Description & UI Elements
1. Executive Dashboard	Dynamic greeting, 4 Live Ring Gauges (Health Score, Exam Readiness, Streak, Today's Progress %), Quick List checklist, AI Motivation.
2. Calendar Habit Tracker	Scoped 35-Day Month Grid, Future-Date Protection (future checkboxes disabled), Goal Clamping (1-31), Daily Rewards Claiming Modal.
3. Wellness Monitor	Screen Time, Sleep, Study, Exercise, Water Stepper (💧), Mood Selector, Stress/Energy/Focus sliders, Health Score Breakdown Gauge.
4. Pomodoro Focus Studio	Circular Timer Ring, Web Audio Synth Beep Engine, Subject Focus Tagger, Study Task Planner, 14-Day Subject Distribution Bar Chart.
5. Performance Analytics	365-Day Activity Heatmap, 30-Day Area Chart, Radar Chart (7-Day Avg), Habit Consistency Table, 1-Click PDF Exporter.
6. Gamification Engine	10 Level Tiers (Seedling → Legend), 15 Achievement Badges, Daily Missions, Coin Economy, Canvas Particle Confetti.
7. Social Community Feed	Peer Leaderboard, Live Activity Stream, Peer Badges & Milestone Feed.



4. Real-World Application & Impact

A. Intended Use Cases

SystemOS is specifically crafted for individuals willing to improve their lives by analyzing self-data, discovering personal flaws, and enforcing daily consistency:

- **Competitive Examination Aspirants (JEE, NEET, UPSC, GATE):** Students managing 8–10 hour study routines, tracking sleep stability, and allocating subject study time without experiencing burnout.
- **Corporate Professionals & Remote Engineers:** Knowledge workers balancing project milestones with screen time boundaries, hydration, and exercise goals.
- **Self-Directed Learners & Habit Builders:** Individuals establishing daily coding, reading, or mindfulness habits with structured gamified rewards.

B. How SystemOS Solves the Core Problem

1. **Demotivates Bad Habits:** Explicit screen time warnings, failure prediction alerts, and stress penalty scoring highlight negative behaviors before habits fail.
2. **Motivates Consistent Good Habits:** Instant XP rewards, level titles, flame streaks, and daily mission coins build strong positive momentum.
3. **Healthy Competition via Leaderboard:** Peer standings and activity feeds induce positive social accountability and constructive regret over missed goals.
4. **100% Privacy Preservation:** Local Ollama LLM execution guarantees sensitive health and routine data remains strictly on-device.

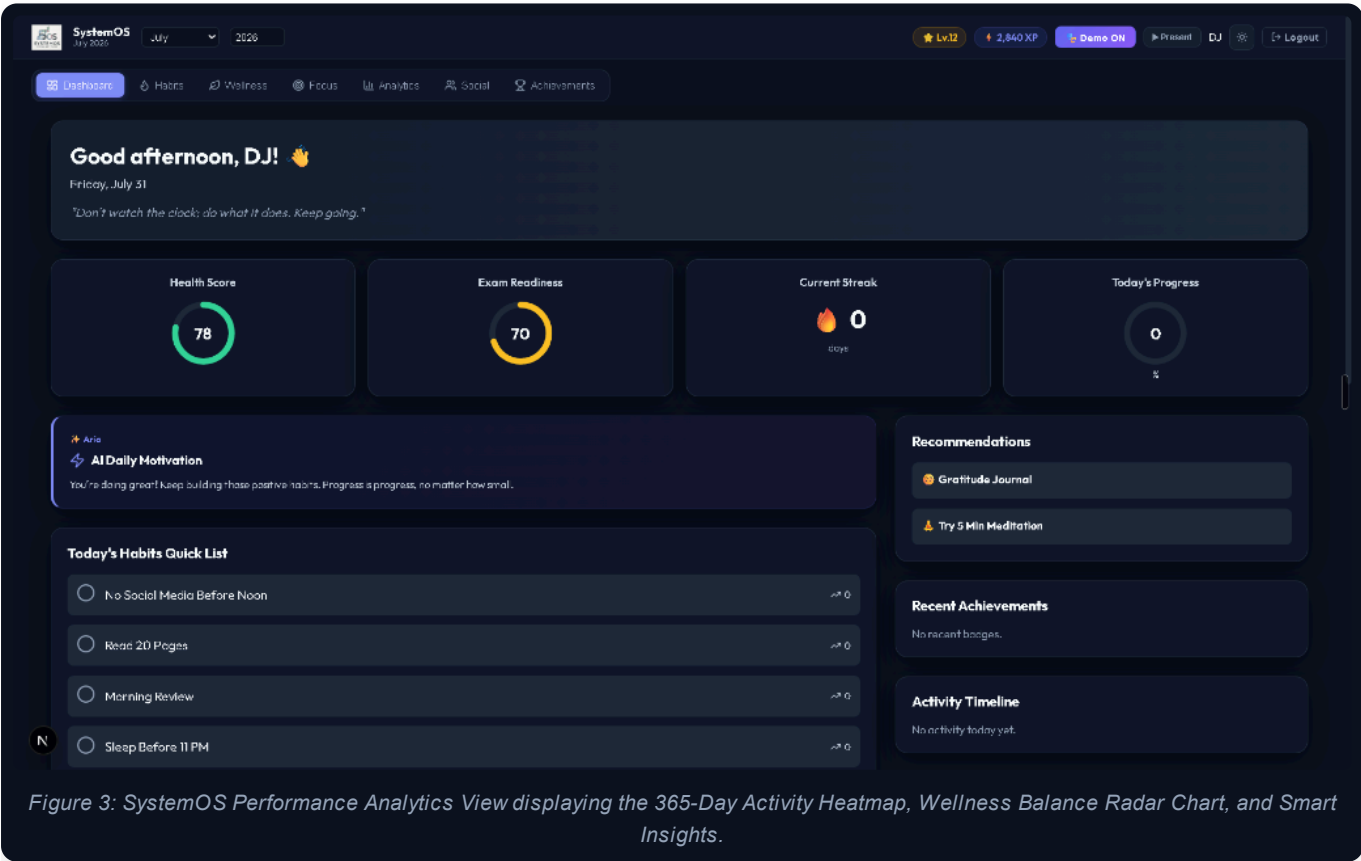


Figure 3: SystemOS Performance Analytics View displaying the 365-Day Activity Heatmap, Wellness Balance Radar Chart, and Smart Insights.

5. Technical Specifications & Formulations

A. Core Mathematical Formulations

Student Health Score Formula (0–100 pts):
Health Score = Sleep(25) + Habits(20) + ScreenTime⁻¹(15) + Hydration(15) + Exercise(10) + Study(10) + Stress⁻¹(5)

Exam Readiness Score Formula (0–100 pts):
Exam Readiness = StudyHoursAvg(35) + SleepConsistency(20) + ScreenPenalty(20) + Pomodoros/10(15) + HabitPct(10)

B. System Architecture Summary

Layer	Technologies Used
Frontend Framework	Next.js 16 (App Router), React 19, Lucide Icons, Recharts
AI Engine	On-Device Ollama ('qwen3:4b'), 4s AbortController Timeout, Rule Engine Fallback
Backend & Database	Prisma ORM, SQLite / MySQL, LocalStorage Zero-Latency Cache
Styling & Design	CSS Modules, Japandi Light Theme, Deep Space Navy Dark Mode, Glassmorphism

6. Conclusion

SystemOS successfully fulfills the vision of an intelligent, privacy-first habit tracker and digital wellness monitor. By integrating analytical graphs, a single-view GUI, peer leaderboards, gamified rewards, and local AI coaching, SystemOS helps students and professionals eliminate distractions, correct personal mistakes, and achieve sustainable daily consistency.

SystemOS Final Submission Report
Generated for Final Hackathon Evaluation | Privacy-First Habit & Digital Health OS